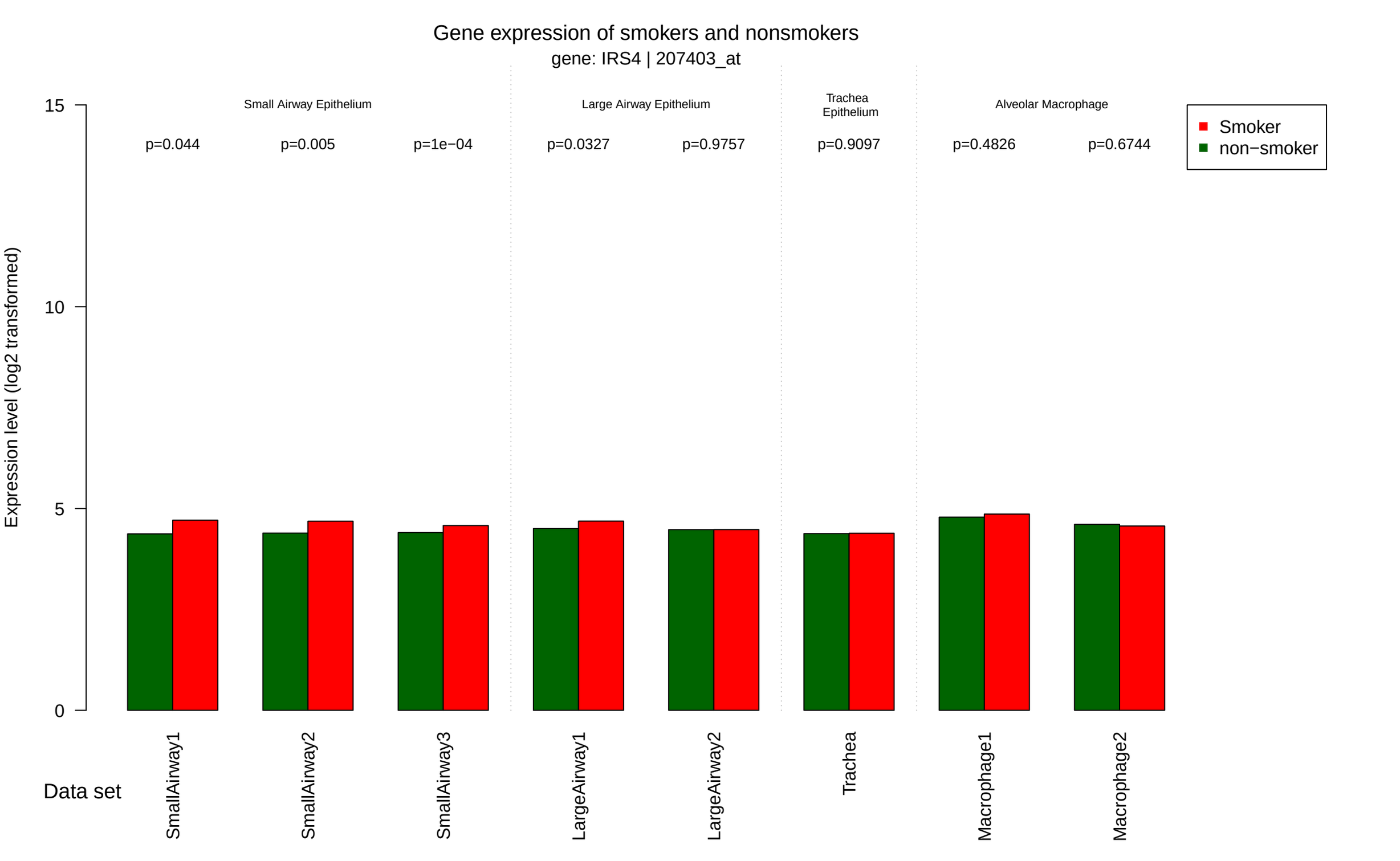
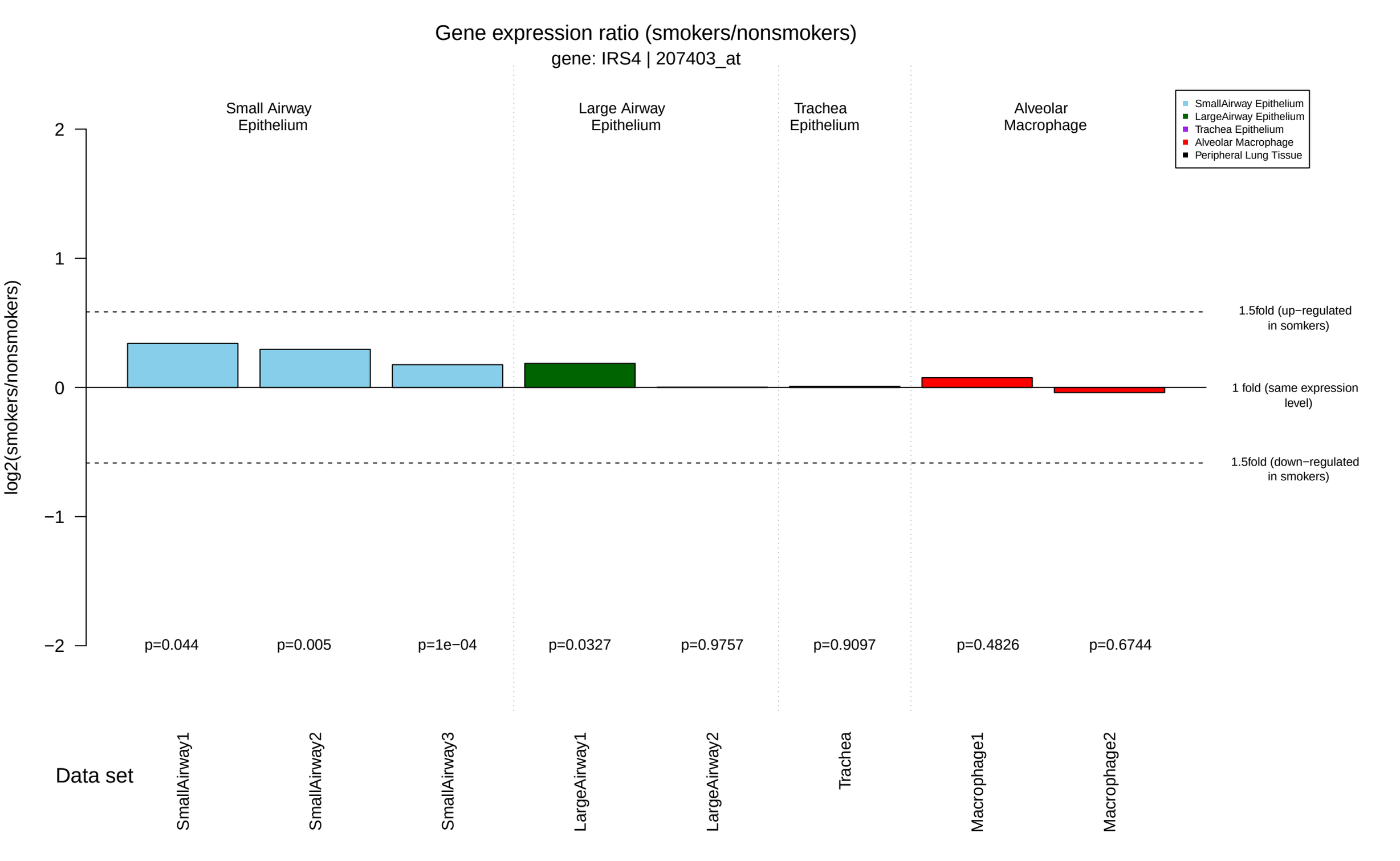
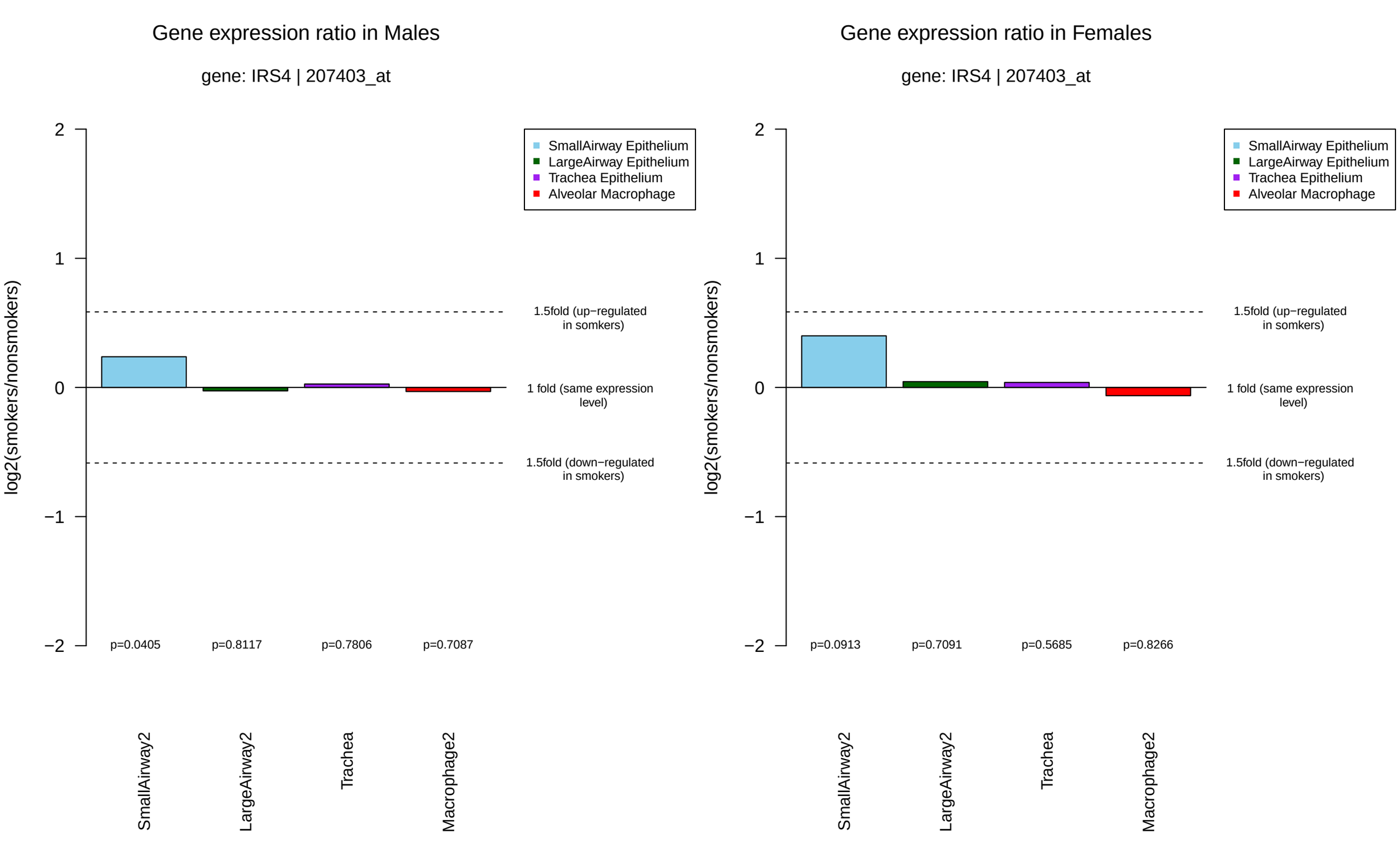
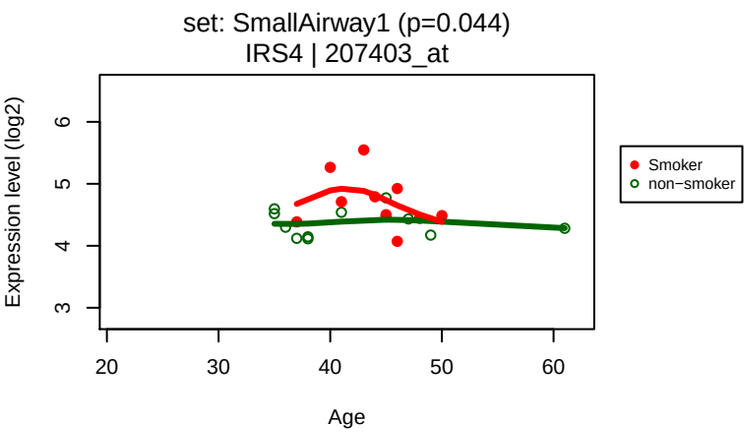
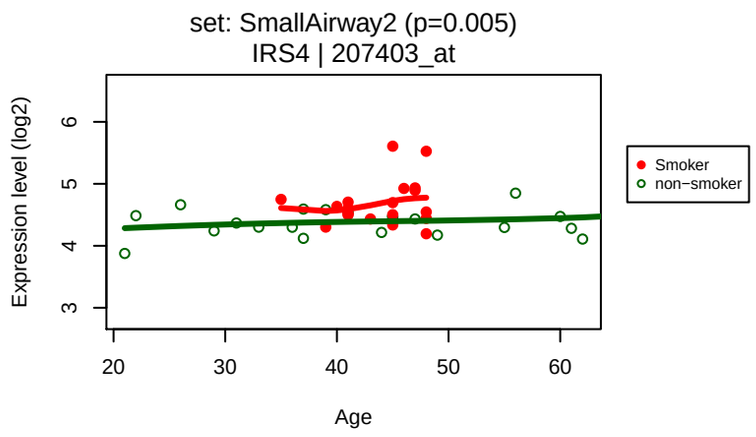
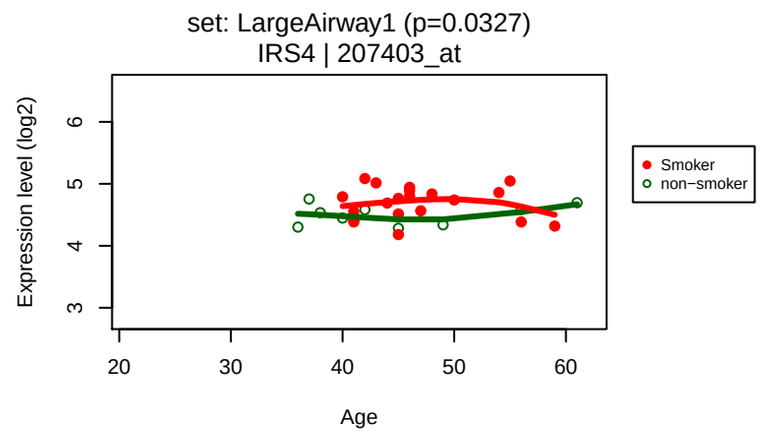

Gene expression of smokers and nonsmokers
gene: IRS4 | 207403_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.044
p=0.005
p=1e−04
p=0.0327
p=0.9757
p=0.9097
p=0.4826
p=0.6744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

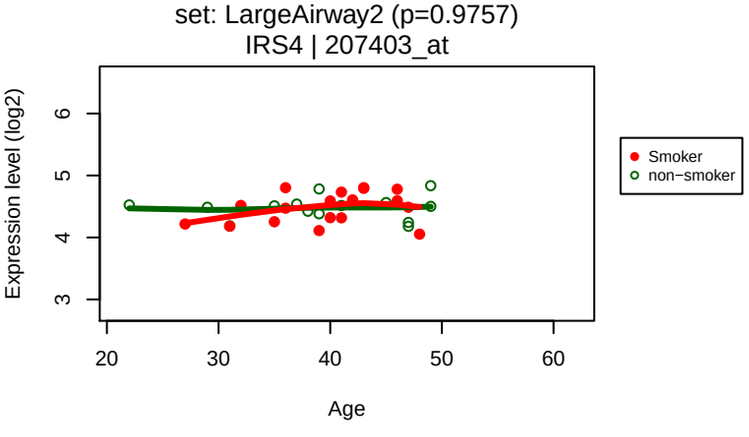

Gene expression ratio (smokers/nonsmokers)
gene: IRS4 | 207403_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.044
p=0.005
p=1e−04
p=0.0327
p=0.9757
p=0.9097
p=0.4826
p=0.6744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

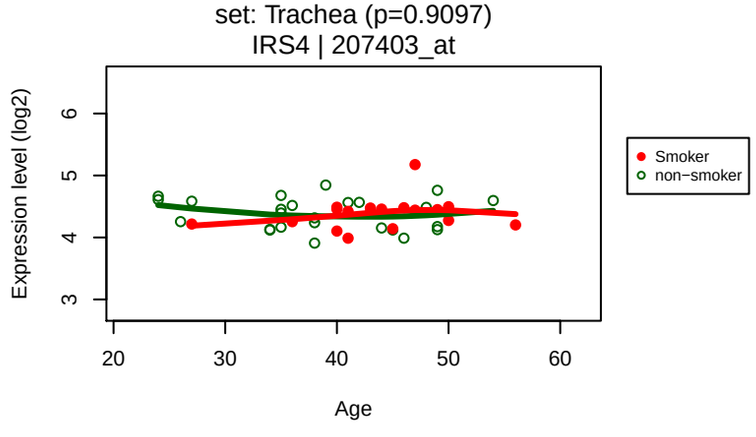

Gene expression ratio in Males
gene: IRS4 | 207403_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0405
p=0.8117
p=0.7806
p=0.7087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRS4 | 207403_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0913
p=0.7091
p=0.5685
p=0.8266
SmallAirway2
LargeAirway2
Trachea
Macrophage2

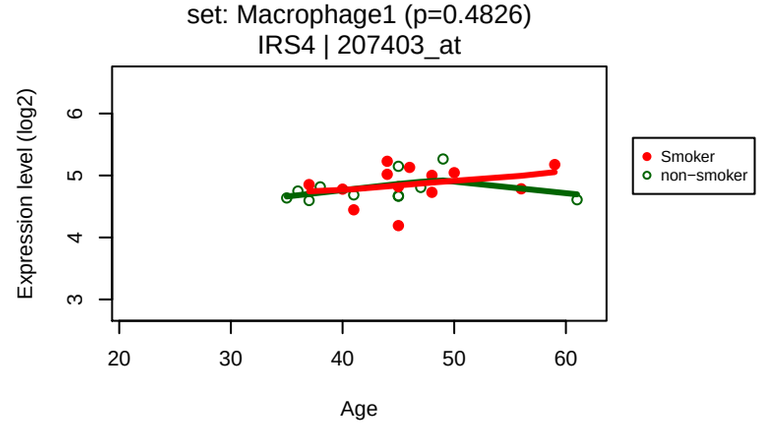

set: SmallAirway1 (p=0.044)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker

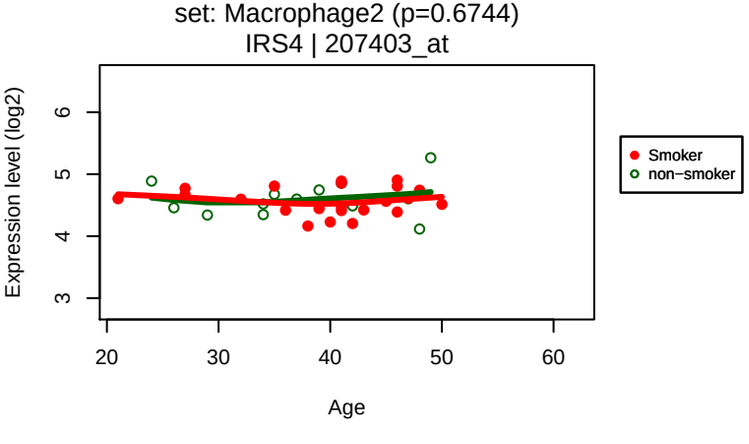

set: SmallAirway2 (p=0.005)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker

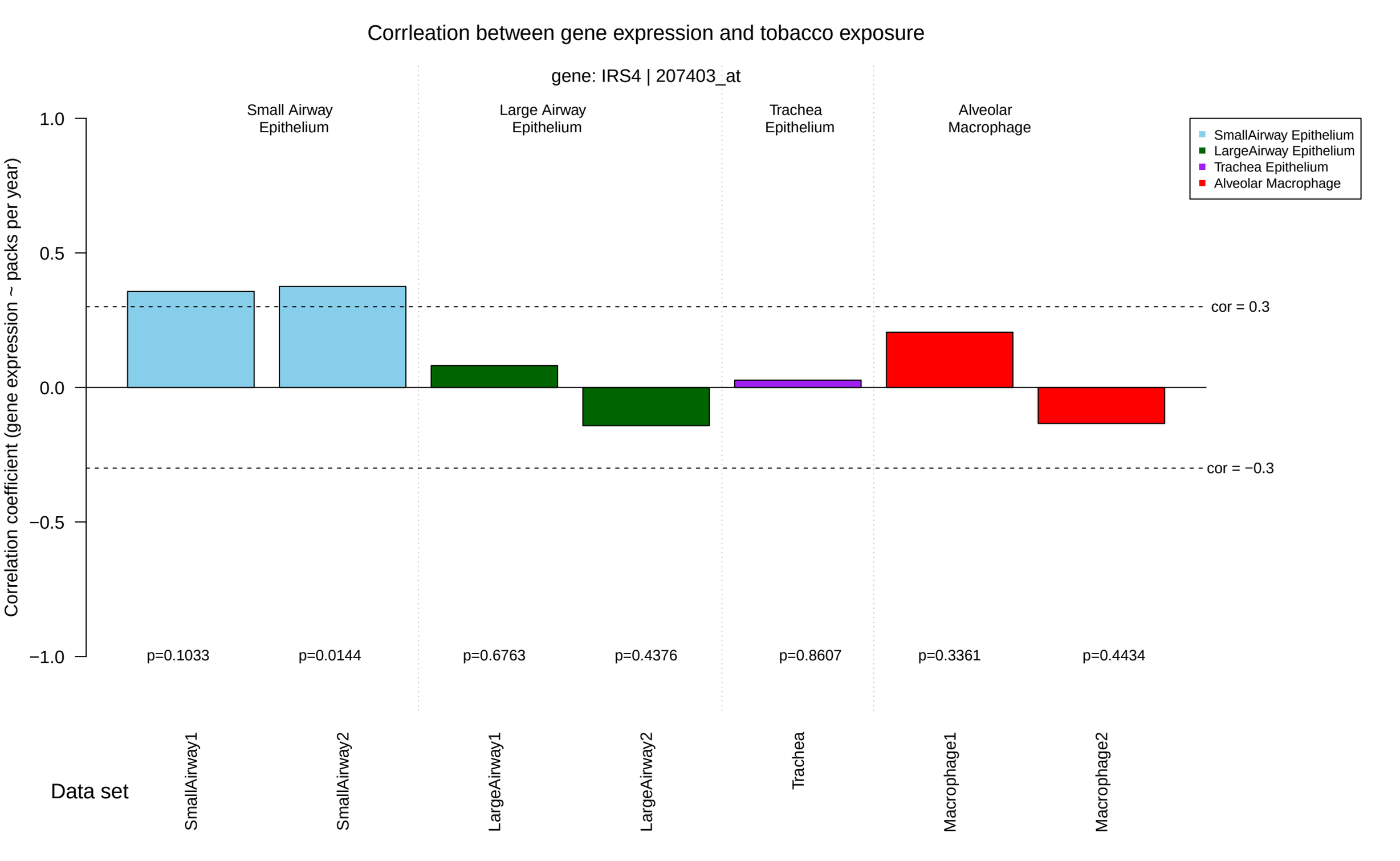

set: LargeAirway1 (p=0.0327)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9757)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9097)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4826)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6744)
IRS4 | 207403_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRS4 | 207403_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1033
p=0.0144
p=0.6763
p=0.4376
p=0.8607
p=0.3361
p=0.4434
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2